

Monitor your hotspot

Optimise the quality of your home Wi-Fi hotspot

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The Wi-Fi access hotspot in your home is one of the most important factors in making sure that you get a great experience on all your devices. The Wi-Fi network is implemented using the IEEE 802.11 protocols, which has variants depending on the frequency used for the wireless data transmission. Wi-Fi routers use one of two frequency bands, 802.11b variant is for the 2.4GHz band while the 802.11a

which is essentially smaller bands over which the data is transmitted. On the 802.11b variant, 14 bands are available all around the world, even though some countries may make fewer bands available. In the US for example, only 11 bands are used. Here in India, all the 14 channels are available.

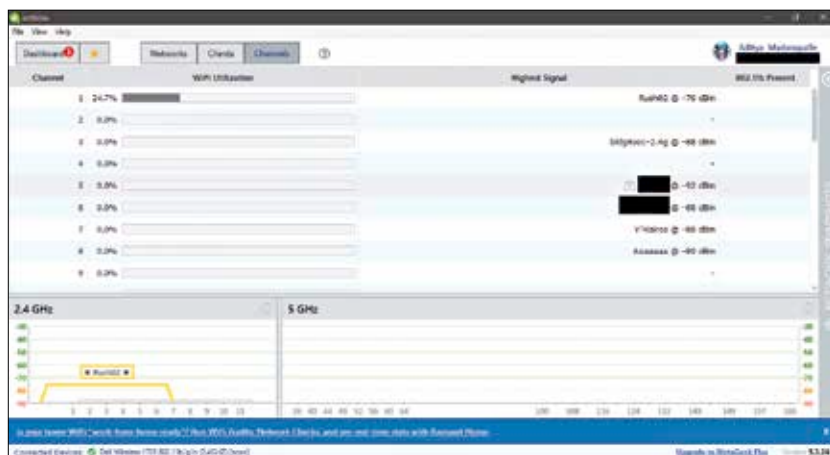
Now why are we worried about channels and how does that affect the user experience? There are three types of interference that are experienced when using Wi-Fi networks. The interference caused by multiple devices connecting to the same access point is called co-channel interference. This can be worked

bluetooth devices and some cameras. The Wi-Fi implementation is robust enough that these sources of interference are rarely problematic. The third source of interference is from devices on adjacent channels, and this is the one that causes the real problem. This interference can be reduced or removed entirely by selecting the best channel.

SELECTING THE BEST CHANNEL

The 14 channels are numbered from 1 to 14, and each channel has a width of 22MHz. The channels are separated by 5MHz, which means that every 5MHz a new channel is assigned. That means that four other channels use some of the medium used by channel 1, which can lead to problems for devices on that channel. For this reason, the clean channels with the least overlap are channels 1, 6 and 11.

Most routers already pick 1, 6 and 11 by default. If you are making a choice here in your device settings, it is safest to go with one of the three options. Now the question is, which of these three is the best for you? That is a question that is tricky to answer. It depends on which Wi-Fi channels are used by the routers of your neighbours. If one neighbour is using channel 6, and another is using channel 11, it makes sense for you to use channel 1. This is putting it in a simple way, but considering the density of Wi-Fi hotspots in modern housing complexes, you might be dealing with multiple SSIDs on each channel, and you will have to check for usage over time to pick an uncrowded part of the medium. Additionally, the location of the device within the household plays a critical role here, because the cluster of SSIDs accessible in the bedroom and the channels they use may be different from the SSIDs accessible in the living room and the channels they use. In fact, there is professional gear to investigate the signal strength of the router across an area, which involves walking around with a specific receiver and mapping out the environment. To do this yourself, all you need is a Wi-Fi diagnostic tool, and there are a number of these included in this month's Alpha DVD. Most of these tools



The user interface of inSSIDer showing a rogue router.

variant is for the 5GHz band. While the 2.4GHz band offers better coverage, the 5GHz band offers better speed, and this is the main difference between the two when it comes to how it affects the end user. The difference is still on the theoretical side, and it is perfectly possible to get great broadband connectivity on the 2.4GHz band as well. In most cases, you will be working with 802.11b in a typical home Wi-Fi setup. There may be up to 45 channels, within a frequency band,

around by choosing dual band routers if needed, but should not really be much of a problem, because each of the frequency bands support a total of 32 devices. Dual band routers can support 64 devices, which is the go to option for an office or business environment. The second source of interference is from non Wi-Fi devices using the same portion of the wireless spectrum, that is non 802.11 devices using the same medium. These can be microwave ovens, cordless phones,